



WINTRIX ACADEMY PRESENTS

DCET Previous Year Question Paper & Solutions

2026

DIPLOMA COMMON ENTRANCE TEST — SET D1

GUIDANCE • STRATEGY • SUCCESS

100 Questions • Complete Answer Key • Subject-Wise Organisation

FIRST EDITION — 2026



DCET 2026 Previous Year Question Paper & Solutions

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About Wintrix Academy

Wintrix Academy is a premier DCET mentorship platform founded with a singular mission: to empower Diploma students across Karnataka with guidance, strategy, and resources to secure top ranks.



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48

DCET RANK

9.8

DIPLOMA
CGPA

9.19

RVCE CGPA

50+

STUDENTS



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Turning confusion into clarity and effort into results through new-age learning strategies and results-engineered systems.

Preface

Dear DCET Aspirant,

This book contains the complete DCET 2026 Question Paper (Set D1) with answers and explanations for all 100 questions, organised subject-wise across five chapters.

The 2026 paper covers Engineering Mathematics (20 marks), Project Management (20 marks), Applied Statistics & Data Analytics (20 marks), Information Technology (20 marks), and Electrical & Electronics Engineering (20 marks) — each subject carrying equal weightage.

Attempt each question on your own first, then verify with the answer and explanation. Focus on understanding concepts rather than memorising answers. Wishing you the very best!

Mohammed Adnan

Founder, Wintrix Academy

Table of Contents

PRELIMINARIES

About Wintrix Academy	iii
Preface	iv

DCET 2026 — QUESTION PAPER & SOLUTIONS

Chapter 1: Mathematics	
<i>Questions 1–20 • 20 Marks</i>	
Chapter 2: Project Management	
<i>Questions 21–40 • 20 Marks</i>	
Chapter 3: Applied Statistics & Data Analytics	
<i>Questions 41–60 • 20 Marks</i>	
Chapter 4: Information Technology	
<i>Questions 61–80 • 20 Marks</i>	
Chapter 5: Electrical & Electronics Engineering	
<i>Questions 81–100 • 20 Marks</i>	

01

CHAPTER 1

Mathematics

20 Questions · 20 Marks

DCET 2026 — Set D1

Q1

MATH

If A is a square matrix given by $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, the principal diagonal elements of above matrix are

(1) 1 2 3

(2) 4 5 6

(3) 1 5 9

(4) 7 5 3

✓ ANSWER: (3)

The principal diagonal of a matrix runs from top-left to bottom-right. For the given 3×3 matrix, the diagonal elements are $a_{11} = 1, a_{22} = 5, a_{33} = 9$.

Q2

MATH

The value of $\begin{vmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{vmatrix}$ is

(1) -1

(2) 0

(3) 1

(4) $\cos 2x$

✓ ANSWER: (3)

Expanding along R_3 : $1 \times (\cos^2 x + \sin^2 x) = 1$

Using the identity $\cos^2 x + \sin^2 x = 1$, the determinant evaluates to 1.

For any square matrix A, identify the correct statement

(1) $A \cdot \text{adj } A = |A|I$

(2) $A \cdot \text{adj } A = 0$

(3) $\text{adj } A = |A|I$

(4) $A \cdot \text{adj } A = 1$

✓ ANSWER: (1)

The standard identity for any square matrix: $A \cdot \text{adj}(A) = |A| \cdot I$, where $\text{adj}(A)$ is the adjugate matrix and I is the identity matrix.

The minors of the elements 5 & 2 in the matrix $A = \begin{bmatrix} 1 & -2 & 4 \\ 5 & 2 & -1 \\ 7 & 6 & 9 \end{bmatrix}$ are

$\begin{vmatrix} 5 & 2 & -1 \end{vmatrix}$

$\begin{bmatrix} 7 & 6 & 9 \end{bmatrix}$ are

(1) 24, 52

(2) -42, -19

(3) -6, 2

(4) 42, 19

✓ ANSWER: (2)

Minor of 5 = $\begin{vmatrix} -2 & 4 \\ 6 & 9 \end{vmatrix} = (-2)(9) - (4)(6) = -18 - 24 = -42$

Minor of 2 = $\begin{vmatrix} 1 & 4 \\ 7 & 9 \end{vmatrix} = (1)(9) - (4)(7) = 9 - 28 = -19$

Q5

MATH

If $x + y + \lambda = 0$ is an equation of a line passing through the point $(1, 2)$, then the value of λ is _____

(1) 3

(2) $1/3$

(3) -3

(4) $-1/3$

✓ ANSWER: (3)

Substituting $(1, 2)$: $1 + 2 + \lambda = 0$

$3 + \lambda = 0 \rightarrow \lambda = -3$

Q6

MATH

If $4y - 3x - 7 = 0$ is a given equation of a straight line, then its slope is _____

(1) $4/3$ (2) $-4/3$ (3) $3/4$ (4) $-3/4$

✓ ANSWER: (3)

$4y - 3x - 7 = 0$

$4y = 3x + 7$

$y = (3/4)x + 7/4$

Slope $m = 3/4$

Q7

MATH

If a straight line is parallel to Y-axis, then its slope is _____

(1) 0

(2) ∞

(3) 1

(4) -1

✓ ANSWER: (2)

A line parallel to the Y-axis is vertical. The slope of a vertical line is **undefined (∞)**, as the change in x is zero.

Equation of a straight line whose angle of inclination with x-axis is 135° and having 4 units as its y-intercept is _____

(1) $x - y - 4 = 0$

(2) $x - y + 4 = 0$

(3) $x + y + 4 = 0$

(4) $x + y - 4 = 0$

✓ ANSWER: (4)

Slope $m = \tan(135^\circ) = -1$

y-intercept $c = 4$

$y = -x + 4 \rightarrow x + y - 4 = 0$

If $y = 2/(1 - x)$, then dy/dx is

(1) 0

(2) $2/(1 - x)^2$

(3) $1/(1 - x)^2$

(4) $-2/(1 - x)^2$

✓ ANSWER: (2)

$y = 2(1 - x)^{-1}$

$dy/dx = 2 \times (-1)(1 - x)^{-2} \times (-1)$

$dy/dx = 2/(1 - x)^2$

The derivative of $e^x \cos x$ is

(1) $e^x (\sin x - \cos x)$

(2) $e^x (-\sin x - \cos x)$

(3) $e^x (-\sin x + \cos x)$

(4) $-e^x (-\sin x + \cos x)$

✓ ANSWER: (3)

$$\begin{aligned} \frac{d}{dx} [e^x \cdot \cos x] &= e^x \cdot (-\sin x) + \cos x \cdot e^x \\ &= e^x (\cos x - \sin x) \\ &= e^x (-\sin x + \cos x) \end{aligned}$$

If $y = e^{3x} + e^{-5x}$ then d^2y/dx^2 at $x = 0$ is

(1) 35

(2) 34

(3) 36

(4) -35

✓ ANSWER: (2)

$$\begin{aligned} \frac{dy}{dx} &= 3e^{3x} - 5e^{-5x} \\ \frac{d^2y}{dx^2} &= 9e^{3x} + 25e^{-5x} \\ \text{At } x = 0: 9(1) + 25(1) &= 34 \end{aligned}$$

Rate of change of area 'A' of a circle with respect to its radius 'r' is

(1) $dA/dr = 2\pi r$

(2) $dA/dr = \pi r^2$

(3) $dA/dr = 2\pi$

(4) $dA/dr = \pi r$

✓ ANSWER: (1)

$$A = \pi r^2$$

$$dA/dr = 2\pi r$$

Match List – I (function) with List – II (integrand) and select the correct answer:

(a) $\int e^x dx \rightarrow$ (i) $\log x + c$

(b) $\int (1/x) dx \rightarrow$ (ii) $e^x + c$

(c) $\int dx \rightarrow$ (iii) $\tan x + c$

(d) $\int \sec^2 x dx \rightarrow$ (iv) $x + c$

(1) a – i, b – ii, c – iv, d – iii

(2) a – ii, b – iii, c – i, d – iv

(3) a – ii, b – i, c – iv, d – iii

(4) a – iv, b – iii, c – ii, d – i

✓ ANSWER: (3)

$\int e^x dx = e^x + c \rightarrow$ a-ii; $\int (1/x) dx = \log x + c \rightarrow$ b-i; $\int dx = x + c \rightarrow$ c-iv; $\int \sec^2 x dx = \tan x + c \rightarrow$ d-iii.

$$\int \sin^6 x \cdot \cos x \, dx$$

(1) $\cos^7 x / 7 + c$

(2) $-\cos^7 x / 7 + c$

(3) $\sin^7 x / 7 + c$

(4) $6 \sin^6 x + c$

✓ ANSWER: (3)

Let $u = \sin x$, $du = \cos x \, dx$
 $\int u^6 \, du = u^7 / 7 + c = \sin^7 x / 7 + c$

The integral of $x \cos x$ with respect to 'x' is

(1) $x \cos x + \sin x + c$

(2) $x \sin x + \cos x + c$

(3) $\cos x + \sin x + c$

(4) $x(\sin x + \cos x) + c$

✓ ANSWER: (2)

Using integration by parts: $\int x \cdot \cos x \, dx$
 $u = x$, $dv = \cos x \, dx$
 $= x \cdot \sin x - \int \sin x \, dx$
 $= x \sin x + \cos x + c$

Q16

MATH

The volume of solid generated by revolving the curve $y = \sqrt{x^2 + 5x}$ between $x = 1$ and $x = 2$ is

(1) $59\pi/6$ cubic units(2) $59/6$ cubic units(3) $58\pi/5$ cubic units(4) $59\pi/5$ cubic units

✓ ANSWER: (1)

$$\begin{aligned}
 V &= \pi \int_1^2 y^2 dx = \pi \int_1^2 (x^2 + 5x) dx \\
 &= \pi [x^3/3 + 5x^2/2]_1^2 \\
 &= \pi [(8/3 + 10) - (1/3 + 5/2)] \\
 &= \pi [38/3 - 11/6] = \pi \times 59/6 = 59\pi/6
 \end{aligned}$$

Q17

MATH

The value of $\sin 480^\circ$ is

(1) $\sqrt{3}/2$ (2) $\sqrt{2/3}$ (3) $\sqrt{3}$ (4) $\sqrt{3}/2 \times 2$

✓ ANSWER: (1)

$$\begin{aligned}
 \sin 480^\circ &= \sin(480^\circ - 360^\circ) = \sin 120^\circ \\
 &= \sin(180^\circ - 60^\circ) = \sin 60^\circ = \sqrt{3}/2
 \end{aligned}$$

Q18

MATH

$\sin(A + B)$ is equal to

(1) $\sin A \cdot \cos B - \cos A \cdot \sin B$ (2) $\sin A \cdot \cos B + \cos A \cdot \sin B$ (3) $\cos A \cdot \cos B + \sin A \cdot \sin B$ (4) $\cos A \cdot \cos B - \sin A \cdot \sin B$

✓ ANSWER: (2)

The compound angle formula: $\sin(A + B) = \sin A \cdot \cos B + \cos A \cdot \sin B$.

If $\cos A = 12/13$ and $\sin B = 3/5$, and A and B are acute angles, the value of $\cos(A - B)$ is

(1) $56/65$

(2) $-16/65$

(3) $33/65$

(4) $63/65$

✓ ANSWER: (4)

$$\begin{aligned}\cos A &= 12/13 \rightarrow \sin A = 5/13 \\ \sin B &= 3/5 \rightarrow \cos B = 4/5 \\ \cos(A-B) &= \cos A \cdot \cos B + \sin A \cdot \sin B \\ &= (12/13)(4/5) + (5/13)(3/5) \\ &= 48/65 + 15/65 = 63/65\end{aligned}$$

The value of $\sin 125^\circ \times \sin 55^\circ$ is equal to

(1) $\sin^2 35^\circ$

(2) $\cos^2 35^\circ$

(3) $\sin 35^\circ$

(4) $\cos 35^\circ$

✓ ANSWER: (2)

$$\begin{aligned}\sin 125^\circ &= \sin(180^\circ - 55^\circ) = \sin 55^\circ = \cos 35^\circ \\ \therefore \sin 125^\circ \times \sin 55^\circ &= \cos 35^\circ \times \cos 35^\circ = \cos^2 35^\circ\end{aligned}$$

02

CHAPTER 2

Project Management

20 Questions · 20 Marks

DCET 2026 — Set D1

Q21

PM

Complete the following Analogy:

Project : Temporary Endeavour :: Operation : _____

(1) Ongoing work effort

(2) Plan

(3) Risk

(4) Deliverables

✓ ANSWER: (1)

A project is a **temporary endeavour**, whereas an operation is an **ongoing work effort** — it is continuous and repetitive, unlike the finite nature of a project.

Q22

PM

Who provides guidance as well as direction to the projects?

(1) Manager

(2) Operator

(3) Consultants

(4) Administrator

✓ ANSWER: (3)

As per the answer key, **Consultants** provide guidance as well as direction to projects, offering expert advice on planning, execution and strategy.

Match the following project-based on their classification and select the correct option.

List – A List – B

- (a) Basis of sector → (i) Private sector project
(b) Basis of ownership → (ii) High risk project
(c) Basis of dependency → (iii) Industrial project
(d) Basis of risk involved in project → (iv) Dependent project

(1) a – iii, b – i, c – iv, d – ii

(2) a – iv, b – i, c – iii, d – ii

(3) a – iii, b – ii, c – iv, d – i

(4) a – ii, b – i, c – iv, d – iii

✓ ANSWER: (2)

Basis of sector → (iv), Basis of ownership → (i) Private sector, Basis of dependency → (iii) Industrial (dependent on industry), Basis of risk → (ii) High risk.

One of the misconceptions about project management is _____.

(1) Planning

(2) Rework

(3) Scheduling

(4) Controlling

✓ ANSWER: (2)

Rework is a common misconception — project management is about proactive planning, not redoing work. Planning, scheduling, and controlling are core PM functions.

Project work system can be designed by developing and preparing _____.

(1) Work breakdown system, Project Execution Plan, Project Procedure Manual

(2) Work breakdown system, Advance Action, Project Diary

(3) Work breakdown system, Project Execution Plan, Project Diary

(4) Work breakdown system, Project Execution Plan, Advance Action

✓ ANSWER: (1)

A project work system is designed using Work Breakdown System, Project Execution Plan, and Project Procedure Manual — these three form the backbone of project planning.

Choose the correct statements for successful project implementation:

(i) Adequate formulation

(ii) Sound project organisation

(iii) Proper implementation planning

(iv) Advance Action

(1) (i) and (ii) are correct

(2) (i), (ii) and (iii) are correct

(3) (ii) and (iii) are correct

(4) All are correct

✓ ANSWER: (4)

All four — adequate formulation, sound organisation, proper implementation planning, and advance action — are essential for successful project implementation.

In any project, who will maintain project diary?

(1) Financier

(2) Administrator

(3) Project Manager

(4) Accountant

✓ ANSWER: (3)

The **Project Manager** is responsible for maintaining the project diary, which records daily activities, progress, and decisions throughout the project lifecycle.

Match the following and choose the correct answer.

List – I List – II

(a) Initial project team → (i) People who invested in the project

(b) Core project team → (ii) Group of people who give feedback and counselling

(c) Project Advisors → (iii) Specific people who initially conceive the idea

(d) Project stakeholders → (iv) Group ultimately responsible for designing and managing project

(1) a – ii, b – iii, c – i, d – iv

(2) a – ii, b – iii, c – iv, d – i

(3) a – iii, b – iv, c – ii, d – i

(4) a – i, b – iii, c – iv, d – ii

✓ ANSWER: (3)

Initial project team (a) → (iii) people who conceive the idea; Core project team (b) → (iv) responsible for designing/managing; Project Advisors (c) → (ii) give feedback; Stakeholders (d) → (i) investors.

Decision tree analysis in project management is used to evaluate _____ outcomes.

(1) Uncertainty

(2) Certainty

(3) Both (1) & (2)

(4) None of the above

✓ ANSWER: (3)

Decision tree analysis evaluates outcomes under both **uncertainty** and **certainty** conditions, helping project managers make informed decisions by mapping out possible scenarios.

What does the letter 'C' stand for in CPM?

- (1) Crash
- (2) Critical
- (3) Cost
- (4) Control

✓ ANSWER: (2)

CPM stands for **Critical Path Method**. The 'C' stands for Critical — a technique used to identify the longest path of activities in a project network.

In a network diagram, critical path is defined as _____

- (1) Longest sequence of dependent activities with shortest project duration
- (2) Longest sequence of dependent activities with longest project duration
- (3) Shortest sequence of dependent activities with shortest project duration
- (4) Shortest sequence of dependent activities with longest project duration

✓ ANSWER: (1)

The **critical path** is the longest sequence of dependent activities that determines the minimum (shortest) total project duration. Any delay on this path delays the entire project.

Match the project plan with the List.

List – I List – II

- (a) Resource plan → (i) Human risk
 (b) Financial plan → (ii) Customer requirements
 (c) Quality plan → (iii) Labour cost
 (d) Risk plan → (iv) Type of labour

(1) a – i, b – ii, c – iii, d – iv

(2) a – ii, b – i, c – iv, d – iii

(3) a – iv, b – iii, c – ii, d – i

(4) a – iii, b – iv, c – i, d – ii

✓ ANSWER: (3)

Resource plan → (iv) Type of labour; Financial plan → (iii) Labour cost; Quality plan → (ii) Customer requirements; Risk plan → (i) Human risk.

Q33

Project planning involves _____

(1) Formal planning only

(2) Informal planning only

(3) Both (1) and (2)

(4) Neither (1) nor (2)

✓ ANSWER: (1)

Project planning primarily involves **formal planning** — documented plans including schedules, budgets, resource allocation, and risk management strategies.

Q34

Complex projects : Linear programming :: Large scale scheduling : _____

(1) Bar chart

(2) Network technique

(3) Gantt chart

(4) Heuristic programming

✓ ANSWER: (2)

Just as linear programming is used for complex projects, **Network technique** (like PERT/CPM) is used for large-scale scheduling to manage complex interdependencies.

Given List – I & List – II, match the following:

List – I List – II

- (a) Gantt chart → (i) Fix/set time target
- (b) Network technique → (ii) Estimate time
- (c) Project design → (iii) Interrelation activities
- (d) Time Estimate → (iv) Blue print for action-oriented activity

(1) a – iii, b – ii, c – iv, d – i

(2) a – ii, b – iii, c – iv, d – i

(3) a – iii, b – ii, c – i, d – iv

(4) a – ii, b – iii, c – i, d – iv

✓ ANSWER: (2)

Gantt chart → (ii) Estimate time; Network technique → (iii) Interrelation activities; Project design → (iv) Blueprint for action; Time Estimate → (i) Fix/set time target.

Methodologies involved in planning is/are:

- (i) Planning by incentive
- (ii) Planning by direction

(1) (i) Only

(2) (ii) Only

(3) (i) and (ii)

(4) neither (i) nor (ii)

✓ ANSWER: (3)

Both **planning by incentive** and **planning by direction** are methodologies involved in project planning.

An audit of a project after it has been commissioned is known as _____

- (1) Performance evaluation (2) Abandonment analysis
(3) Post-audit (4) Initial review

✓ ANSWER: (3)

A **Post-audit** is an evaluation conducted after a project is completed and commissioned, to assess whether objectives were achieved and learn for future projects.

Which network technique is suitable for uncertain projects?

- (1) Construction network (2) PERT
(3) CPM (4) Milestone chart

✓ ANSWER: (2)

PERT (Program Evaluation and Review Technique) is suitable for uncertain projects as it uses probabilistic time estimates (optimistic, most likely, pessimistic) to handle uncertainty.

Match the following and select the correct answer:

List – I List – II

- (a) IOT → (i) Gmail account
(b) Augmented reality → (ii) Smart farming
(c) Cloud technology → (iii) Gaming

- (1) a – ii, b – iii, c – i (2) a – ii, b – i, c – iii
(3) a – i, b – ii, c – iii (4) a – iii, b – ii, c – i

✓ ANSWER: (1)

IoT → (ii) Smart farming (sensors in agriculture); Augmented reality → (iii) Gaming (AR overlays in games); Cloud technology → (i) Gmail account (cloud-based service).

The critical path in the given network diagram is _____

[Network: Start → A → B → C → E → End, A → C, A → D → E]

(1) A – B – E

(2) A – B – C – E

(3) A – D – E

(4) A – C – B – E

✓ ANSWER: (2)

The critical path A → B → C → E has the longest duration among all paths, making it the critical (longest) path that determines minimum project completion time.

03

CHAPTER 3

Applied Statistics & Data Analytics

20 Questions · 20 Marks

DCET 2026 — Set D1

Q41

STATS

_____ is a method for graphically depicting groups of numerical data through their quartiles.

(1) Leaf – Plot

(2) Box – Plot

(3) Line – Plot

(4) Pie – Chart

✓ ANSWER: (2)

A **Box Plot** (box-and-whisker plot) displays data distribution using five-number summary: minimum, Q1, median, Q3, and maximum.

Q42

STATS

Which of the following is true with respect to histogram?

(1) Histogram is represented for ordinal data set

(2) In histogram width is equal to the class interval and height is equal to relative frequency

(3) Histogram is used to represent continuous data which is equal to the width of class interval

(4) Histogram is used to represent the nominal data

✓ ANSWER: (3)

A **histogram** is used to represent continuous data where the width of each bar equals the class interval. It shows the frequency distribution of continuous variables.

Q43

STATS

From the bar graph showing food grain production (Rice and Wheat) from 2020 to 2023:
In which years is rice production more than wheat?

- (1) 2021 and 2022 (2) 2020 and 2023
(3) 2020 and 2021 (4) 2022 and 2023

✓ ANSWER: (3)

From the bar graph, rice production (hatched bars) exceeds wheat production (plain bars) in the years **2020 and 2021**.

Q44

STATS

From the same bar graph:
In which two years was an equal number of wheat production observed?

- (1) 2022 and 2023 (2) 2021 and 2023
(3) 2020 and 2021 (4) 2021 and 2022

✓ ANSWER: (2)

From the bar graph, wheat production was approximately equal in **2021 and 2023**.

The number of leaves present in the following stem-leaf plot:

Stem | Leaf

1 | 2, 4, 5, 7

2 | 0, 1, 3

3 | 5, 9

4 | 3, 4, 5

5 | 2, 6, 7, 8, 9

(1) 5

(2) 4

(3) 12

(4) 17

✓ ANSWER: (4)

Stem 1: 4 leaves | Stem 2: 3 leaves | Stem 3: 2 leaves

Stem 4: 3 leaves | Stem 5: 5 leaves

Total = 4 + 3 + 2 + 3 + 5 = 17

The formula to compute the mean for a set of n values $x_1, x_2, \dots, x_n$ is

(1) Mean = $\bar{x} = \sum x_i / n$

(2) Mean = $\bar{x} = \sum x_i / (n + 1)$

(3) Mean = $\bar{x} = (1/x)(\sum x - n)$

(4) Mean = $\bar{x} = 1(\sum x + n) / x$

✓ ANSWER: (1)

The arithmetic mean is calculated as $\bar{x} = \sum x_i / n$ — the sum of all values divided by the total number of observations.

The sum of deviations from mean ($\bar{x}$) is always _____

(1) 1

(2) 0

(3) 2

(4) -1

✓ ANSWER: (2)

A fundamental property of the arithmetic mean: $\sum (x_i - \bar{x}) = 0$. The sum of deviations from the mean is always zero.

For the given set of data which of the following statements are correct?

Statement A: Mean is uniquely defined for a set of data

Statement B: Median does not depend on all the observations

Statement C: Mode is affected by extreme values

(1) A and B

(2) A and C

(3) B and C

(4) A, B and C

✓ ANSWER: (1)

Statement A is correct (mean is unique). Statement B is correct (median depends only on middle values). Statement C is incorrect — **Mode is NOT affected** by extreme values; the mean is.

The formula to compute quartile deviation of a data set is _____

(1) $QD = (Q_2 - Q_1)/2$

(2) $QD = (Q_1 - Q_3)/2$

(3) $QD = (Q_2 + Q_1)/2$

(4) $QD = (Q_3 - Q_1)/2$

✓ ANSWER: (4)

Quartile Deviation (Semi-Interquartile Range) = $QD = (Q_3 - Q_1)/2$, where Q_3 is the third quartile and Q_1 is the first quartile.

Match the following and choose the correct option.

List – I List – II

- (a) Range → (i) Middle number in sorted list of data
(b) Mean → (ii) Most often repeated value in a data set
(c) Mode → (iii) Difference between largest and smallest value of data set
(d) Median → (iv) Average of n observations

(1) a – iv, b – ii, c – i, d – iii

(2) a – iii, b – iv, c – i, d – ii

(3) a – iii, b – iv, c – ii, d – i

(4) a – i, b – iii, c – iv, d – ii

✓ ANSWER: (3)

Range (a) → (iii) Difference between largest and smallest; Mean (b) → (iv) Average; Mode (c) → (ii) Most repeated value; Median (d) → (i) Middle number in sorted list.

Which of the following is not a Python package or library for science and numerical computations?

(1) Numpy

(2) Loops

(3) Matplotlib

(4) Pandas

✓ ANSWER: (2)

Loops is a programming construct (for, while), not a Python package. NumPy, Matplotlib, and Pandas are all popular Python libraries for scientific computing.

Why is indentation important in Python?

(1) For Decoration

(2) To define blocks of code

(3) Increase speed

(4) To define libraries

✓ ANSWER: (2)

In Python, **indentation defines blocks of code** (like loops, functions, conditionals). Unlike other languages that use braces {}, Python uses whitespace indentation as syntax.

In Python programming language, when a loop is used inside another loop, it is called as

(1) For loop

(2) While loop

(3) Nested for loop

(4) Inside loop

✓ ANSWER: (3)

A loop placed inside another loop is called a **Nested loop**. It can be a nested for loop or nested while loop.

Write the output of this programme:

```
a = 5
```

```
b = 8
```

```
n = a**3 + b
```

```
m = n % 2
```

```
print (m)
```

(1) 0

(2) -1

(3) 1

(4) 2

✓ ANSWER: (3)

```
a = 5, b = 8
```

```
n = 53 + 8 = 125 + 8 = 133
```

```
m = 133 % 2 = 1 (remainder when 133 ÷ 2)
```

```
Output: 1
```


Match List – I with List – II and select the correct answer:

- (a) $a = 20 \rightarrow$ (i) Complex number
(b) $b = \text{"Hello Python"} \rightarrow$ (ii) Integer
(c) $c = 12.5 \rightarrow$ (iii) String
(d) $d = 3.0 + 4.1j \rightarrow$ (iv) Float

(1) $a - \text{ii}, b - \text{iii}, c - \text{i}, d - \text{iv}$

(2) $a - \text{iii}, b - \text{iv}, c - \text{ii}, d - \text{i}$

(3) $a - \text{ii}, b - \text{iii}, c - \text{iv}, d - \text{i}$

(4) $a - \text{i}, b - \text{ii}, c - \text{iii}, d - \text{iv}$

✓ ANSWER: (3)

$a = 20 \rightarrow$ (ii) Integer; $b = \text{"Hello Python"} \rightarrow$ (iii) String; $c = 12.5 \rightarrow$ (iv) Float; $d = 3.0 + 4.1j \rightarrow$ (i) Complex number.

What is the output of the given programme?

```
x = 0
```

```
if x == 1:
```

```
    print ("A")
```

```
else:
```

```
    print ("B")
```

(1) B

(2) A

(3) error

(4) No output

✓ ANSWER: (1)

$x = 0$, and $0 \neq 1$

The condition $(x == 1)$ is False

So the else block executes $\rightarrow$ Output: B

Number of students in a class room is an example of _____

- (1) Qualitative data (2) Quantitative data and discrete data
- (3) Ordinal data (4) Quantitative and continuous data

✓ ANSWER: (2)

Number of students is **quantitative** (numerical) and **discrete** (whole numbers — you can't have 25.5 students).

The fastest data collection tool is _____

- (1) Interview (2) Survey
- (3) Google form survey (4) Group discussion

✓ ANSWER: (3)

Google Form survey is the fastest data collection tool as it enables instant distribution to large groups, automatic response collection, and real-time data compilation.

Match List – I (Examples) with List – II (Data types) and select the correct answer:

- (a) Height of a person → (i) Nominal data
- (b) Number of teachers in a school → (ii) Continuous data
- (c) Aadhar Number → (iii) Ordinal data
- (d) Student grades → (iv) Discrete data

- (1) a – ii, b – i, c – iii, d – iv (2) a – iv, b – ii, c – i, d – iii
- (3) a – ii, b – iv, c – i, d – iii (4) a – i, b – iii, c – iv, d – ii

✓ ANSWER: (3)

Height → (ii) Continuous; Number of teachers → (iv) Discrete; Aadhar Number → (i) Nominal (just an ID, no order); Student grades → (iii) Ordinal (A > B > C).

The process of grouping the data according to their characteristics is called _____

(1) Organisation of data

(2) Classification of data

(3) Tabulation of data

(4) Presentation of data

✓ ANSWER: (2)

Classification is the process of arranging data into groups or classes based on their common characteristics or attributes.

04

CHAPTER 4

Information Technology

20 Questions · 20 Marks

DCET 2026 — Set D1

Q61

IT

Find the output of the following program:

```
main ()
{
    int a = 125;
    Printf ("%d\n", 10 + a ++);
}
```

(1) 136

(2) 135

(3) 126

(4) 125

✓ ANSWER: (2)

a = 125
a++ is post-increment: uses current value first, then increments
10 + a++ → 10 + 125 = 135 (a becomes 126 after)
Output: 135

Q62

IT

Which of the following is an example for valid identifier?

(1) \$abcd

(2) 12_34a

(3) /M_ab

(4) abcd

✓ ANSWER: (4)

In C, valid identifiers must start with a letter or underscore. **abcd** is valid. \$abcd (\$ not allowed), 12_34a (starts with digit), /M_ab (/ not allowed) are all invalid.

Match standard symbols (List – I) used in program flowcharts with the purpose of use (List – II).

- (a) Rectangle → (i) Input/Output
- (b) Diamond → (ii) Predefined processing
- (c) Rectangle with lines → (iii) Computer Processing
- (d) Parallelogram → (iv) Decision

(1) a – i, b – iii, c – iv, d – ii

(2) a – iii, b – iv, c – ii, d – i

(3) a – iii, b – ii, c – iv, d – i

(4) a – i, b – iv, c – iii, d – ii

✓ ANSWER: (2)

Rectangle → (iii) Processing; Diamond → (iv) Decision; Rectangle with double lines → (ii) Predefined processing/Subroutine; Parallelogram → (i) Input/Output.

Which of the following is not a valid component of an Android application?

(1) Activity

(2) Service

(3) Application

(4) Communication

✓ ANSWER: (4)

The four main Android components are Activity, Service, Broadcast Receiver, and Content Provider.

Communication is not a valid Android component.

Match the following CSS property to the effect it has on the webpage.

- (a) Padding → (i) Specifies the transparency level of an element
- (b) Margin → (ii) Creates space inside an element, around the content
- (c) Z-index → (iii) Creates space outside an element's border
- (d) Opacity → (iv) Controls the vertical stacking order of elements

(1) a – iii, b – ii, c – iv, d – i

(2) a – ii, b – iii, c – iv, d – i

(3) a – ii, b – iii, c – i, d – iv

(4) a – i, b – iv, c – iii, d – ii

✓ ANSWER: (2)

Padding → (ii) space inside element; Margin → (iii) space outside border; Z-index → (iv) stacking order; Opacity → (i) transparency level.

Which of the following software applications is primarily used to read and render HTML web pages?

(1) Server

(2) Operating system

(3) Web browser

(4) Database

✓ ANSWER: (3)

A **Web browser** (Chrome, Firefox, Safari) is the software application specifically designed to read, interpret, and render HTML web pages.

Give the JavaScript method that removes the last element of an array

(1) Shift ()

(2) POP ()

(3) Slice ()

(4) Splice ()

✓ ANSWER: (2)

pop() removes and returns the last element of an array. **shift()** removes the first element, **slice()** returns a section, and **splice()** adds/removes at any position.

In a standard HTML form submission, what is the correct order of operations?

- (a) Form validation (client-side)
- (b) HTTP Request sent
- (c) User clicks submit
- (d) Server Response/Redirect

(1) a → c → b → d

(2) c → a → b → d

(3) c → b → a → d

(4) a → b → c → d

✓ ANSWER: (2)

The correct order: (c) User clicks submit → (a) Client-side form validation → (b) HTTP Request sent to server → (d) Server processes and sends Response/Redirect.

Match the functional areas of operation (List – I) with business function (List – II) with respect to Enterprise Resource Planning:

- (a) Marketing and sales → (i) Purchasing goods and new materials
- (b) Accounting and Finance → (ii) Recruiting and hiring
- (c) Human Resources → (iii) Marketing a product
- (d) Supply chain management → (iv) Financial accounting of payments from customers and to suppliers

(1) a – i, b – ii, c – iv, d – iii

(2) a – iii, b – iv, c – ii, d – i

(3) a – iii, b – ii, c – iv, d – i

(4) a – ii, b – iii, c – i, d – iv

✓ ANSWER: (2)

Marketing & sales → (iii) Marketing; Accounting & Finance → (iv) Financial accounting; HR → (ii) Recruiting/hiring; Supply chain → (i) Purchasing goods.

Which of the following modules of Enterprise Resource Planning handle payroll and employee related data?

- (1) Supply chain Management (2) Marketing Automation
- (3) Human Resource Management (4) Accounting and Finance

✓ ANSWER: (3)

The **Human Resource Management (HRM)** module handles payroll, employee records, attendance, benefits, and all employee-related data in an ERP system.

Choose the correct sequence of events for any sale with respect to SAP Enterprise resource planning sales.

(i) Pre-sales activities (ii) Inventory sourcing (iii) Billing (iv) Payment (v) Sales order processing (vi) Delivery

- (1) (i), (iii), (iv), (ii), (v), (vi) (2) (i), (iv), (iii), (v), (vi), (ii)
- (3) (i), (v), (ii), (vi), (iii), (iv) (4) (ii), (v), (iii), (vi), (iv), (i)

✓ ANSWER: (3)

SAP sales cycle: (i) Pre-sales → (v) Sales order processing → (ii) Inventory sourcing → (vi) Delivery → (iii) Billing → (iv) Payment.

Which of the following is a part of business process automation technology stack?

- (1) Low-code development (2) Android development
- (3) Software design and development (4) Hardware development

✓ ANSWER: (3)

Software design and development is a core component of business process automation technology stack, enabling creation of automated workflows and business solutions.

Which is the example of SaaS cloud service models?

- (1) AWS Lambda (2) Amazon EC2
(3) Google App Engine (4) Gmail

✓ ANSWER: (4)

Gmail is a SaaS (Software as a Service) product — users access it via browser without managing infrastructure. AWS Lambda = FaaS, EC2 = IaaS, App Engine = PaaS.

Identify which of the following is NOT a characteristic of a public cloud?

- (1) Multi-tenancy (2) High availability and Redundancy
(3) Managed by a third-party provider (4) Dedicated hardware for single user

✓ ANSWER: (4)

Dedicated hardware for single user is a characteristic of a private cloud, not public cloud. Public clouds share resources among multiple tenants.

Match List – I (Types of Cloud services) with List – II (service) and select the correct answer:

- (a) IaaS → (i) Delivers software application over the internet
(b) PaaS → (ii) Allows developers to build applications without managing servers
(c) SaaS → (iii) Provides fundamental computing, network and storage resources
(d) Serverless computing → (iv) Offers a hosted environment for developers to build, test and deploy applications

- (1) a – iv, b – iii, c – ii, d – i (2) a – iii, b – ii, c – i, d – iv
(3) a – iii, b – iv, c – i, d – ii (4) a – i, b – iii, c – ii, d – iv

✓ ANSWER: (3)

IaaS → (iii) fundamental computing resources; PaaS → (iv) hosted environment to build/test/deploy;
SaaS → (i) software over internet; Serverless → (ii) build apps without managing servers.

Match List – I (Cloud deployment models) with List – II (operations) and select the correct answer:

- (a) Public cloud → (i) Designed for shared needs of multiple organizations
- (b) Private cloud → (ii) A combination of public and private clouds
- (c) Hybrid cloud → (iii) Resources are used by a single organization
- (d) Community cloud → (iv) Resources are owned and operated by third-party provider

(1) a – i, b – ii, c – iii, d – iv

(2) a – iv, b – iii, c – ii, d – i

(3) a – iii, b – iv, c – i, d – ii

(4) a – ii, b – iii, c – iv, d – i

✓ ANSWER: (2)

Public cloud → (iv) third-party owned; Private cloud → (iii) single organization; Hybrid → (ii) combination; Community → (i) shared needs of multiple organizations.

What mechanism does a firewall use to decide whether to allow or block network traffic?

(1) The physical location of the device

(2) Pre-defined security rules and criteria

(3) Type of operating system used

(4) Source code of the application

✓ ANSWER: (2)

A firewall uses **pre-defined security rules and criteria** to filter network traffic, allowing or blocking packets based on IP addresses, ports, protocols, and other parameters.

Which of the following is true about spear phishing?

- (1) Spear phishing does not involve malicious links
- (2) Phishing is only used for stealing credit card information
- (3) Spear phishing targets a specific individual or organization
- (4) Spear phishing uses telephone calls instead of emails

✓ ANSWER: (3)

Spear phishing is a targeted attack aimed at a specific individual or organization, unlike general phishing which targets many people randomly.

Match List – I (Cyber threats) with List – II (attacks) and select the correct answer:

- (a) Phishing → (i) Threatening to leak data
- (b) Ransomware → (ii) Overwhelming server to disrupt services
- (c) DoS attack → (iii) Compromising vendors to infiltrate target organization
- (d) Supply chain attack → (iv) Attackers are deceptive emails to reveal credentials

- (1) a – i, b – iii, c – ii, d – iv
- (2) a – iii, b – iv, c – i, d – ii
- (3) a – iv, b – i, c – ii, d – iii
- (4) a – ii, b – iv, c – iii, d – i

✓ ANSWER: (3)

Phishing → (iv) deceptive emails; Ransomware → (i) threatening to leak data; DoS → (ii) overwhelming server; Supply chain → (iii) compromising vendors.

Which type of malware is capable of self-replication and spreading by creating copies of itself through infected systems?

(1) Trojan Horses

(2) Logic Bombs

(3) Spyware

(4) Computer worms

✓ ANSWER: (4)

Computer worms are malware that self-replicate and spread across networks without requiring user interaction or a host program, unlike viruses.

05

CHAPTER 5

Electrical & Electronics Engineering

20 Questions · 20 Marks

DCET 2026 — Set D1

Q81

EEE

Identify the logic gate in which the output is high, when the two inputs are different

(1) AND gate

(2) OR gate

(3) XOR gate

(4) NAND gate

✓ ANSWER: (3)

The XOR (Exclusive OR) gate outputs HIGH (1) only when the two inputs are different. When both inputs are same (0,0 or 1,1), output is LOW (0).

Q82

EEE

Choose the correct answers.

Sensors are used to perform the following functions.

(a) Control the actions as well as drive the plant directly

(b) To measure input signals for feed-forward control

(c) To measure process variables for system monitoring, diagnosis and supervisory control

(d) To generate control signals

(1) (a) and (b)

(2) (b) and (c)

(3) (c) and (d)

(4) (d) and (a)

✓ ANSWER: (2)

Sensors are primarily used to measure input signals for feed-forward control (b) and to measure process variables for system monitoring, diagnosis and supervisory control (c). They do not directly control/drive plants or generate control signals.

PLC is an acronym for _____.

- (1) Pre-Logic Controller (2) Programmable Logic Controller
- (3) Programmable Logic Communication (4) Post-Logic Controller

✓ ANSWER: (2)

PLC stands for **Programmable Logic Controller** — a digital computer used for automation of industrial processes.

Identify the device used for the following conversion:

[i/p signal: AC sine wave → Device → o/p signal: DC pulsating wave]

- (1) Inverter (2) Rectifier
- (3) Cycloconverter (4) Chopper

✓ ANSWER: (2)

A **Rectifier** converts AC (alternating current) to DC (direct current). An inverter does the opposite (DC to AC).

The block diagram with CPU, RAM, ROM, I/O, Timer, and Serial communication port represents the block diagram for

- (1) Microprocessor (2) Computer
- (3) PLC (4) Micro controller

✓ ANSWER: (4)

A **Microcontroller** integrates CPU, RAM, ROM, I/O ports, Timer, and Serial communication port on a single chip, unlike a microprocessor which requires external components.

The material not used in fire extinguishers is _____

- (1) CO₂ (2) Foam
(3) Dry powder (4) Nitrogen

✓ ANSWER: (4)

Common fire extinguisher materials include CO₂, foam, dry powder, and water. **Nitrogen** is an inert gas not used as a fire extinguishing agent.

Pipe material used in earthing is _____

- (1) Nickel & Aluminium (2) Aluminium & Tungsten
(3) Tungsten & Galvanised Iron (4) Galvanised Iron

✓ ANSWER: (4)

Galvanised Iron (GI) pipes are used for earthing because they resist corrosion, provide good conductivity, and are durable when buried in soil.

Which one of the following is the conventional source of Electrical Energy?

- (1) Solar Power (2) Wind Power
(3) Tidal Power (4) Thermal Power

✓ ANSWER: (4)

Thermal Power (coal/gas/oil-based) is a conventional energy source. Solar, wind, and tidal are non-conventional (renewable) energy sources.

Which of the following electrical quantities and their SI units are correctly paired?

- (a) Current — Ampere
- (b) Emf — Volts
- (c) Electrical Energy — Watts
- (d) Resistance — Mho

(1) a, b, c and d

(2) Only a and b

(3) Only c and d

(4) Only b and c

✓ ANSWER: (2)

Current → Ampere ✓ and Emf → Volts ✓ are correct. Electrical Energy is measured in Joules (not Watts — that's Power). Resistance is measured in Ohms (not Mho — that's Conductance).

Match List – I (Electrical Quantity) with List – II (Measuring Instruments):

- (a) Current → (i) Voltmeter
- (b) Potential difference → (ii) Ammeter
- (c) Power → (iii) Ohmmeter
- (d) Resistance → (iv) Watt meter

(1) a – ii, b – i, c – iv, d – iii

(2) a – iv, b – i, c – ii, d – iii

(3) a – iii, b – iv, c – i, d – ii

(4) a – ii, b – iii, c – iv, d – i

✓ ANSWER: (1)

Current → (ii) Ammeter; Potential difference → (i) Voltmeter; Power → (iv) Watt meter; Resistance → (iii) Ohmmeter.

Find the value of current 'I' in the circuit given:

[Circuit: Two 4Ω resistors in parallel, in series with 2Ω , powered by 4V battery]

(1) 2A

(2) 1A

(3) 1.5A

(4) 2.5A

✓ ANSWER: (2)

$$4\Omega \parallel 4\Omega = (4 \times 4) / (4 + 4) = 2\Omega$$

$$\text{Total } R = 2\Omega + 2\Omega = 4\Omega$$

$$I = V/R = 4/4 = 1A$$

For the circuit given, calculate the voltage drop across 3Ω Resistor:

[Circuit: 3Ω in series with ($6\Omega \parallel 6\Omega$), powered by 6V]

(1) 6 V

(2) 3 V

(3) 1 V

(4) 2 V

✓ ANSWER: (2)

$$6\Omega \parallel 6\Omega = (6 \times 6) / (6 + 6) = 3\Omega$$

$$\text{Total } R = 3\Omega + 3\Omega = 6\Omega$$

$$I = V/R = 6/6 = 1A$$

$$\text{Voltage across } 3\Omega = I \times R = 1 \times 3 = 3V$$

The function/s of fuse wire is/are to:

- (a) Carry normal working current without heating
- (b) Break the circuit during short circuit
- (c) Protect the circuit from undervoltage
- (d) Protect the circuit from overvoltage

(1) a, b & c

(2) b only

(3) a & b

(4) a, b & d

✓ ANSWER: (3)

A fuse wire (a) carries normal current without heating and (b) breaks the circuit during short circuit/overcurrent. It does NOT protect against undervoltage or overvoltage.

Rating of a fuse wire is _____

(1) Maximum current at which a fuse burns

(2) Maximum voltage at which the fuse burns

(3) Maximum current which fuse can carry without burning

(4) Minimum current at which fuse burns

✓ ANSWER: (3)

Fuse rating = **maximum current the fuse can safely carry without blowing/burning**. If current exceeds this rating, the fuse melts and breaks the circuit.

Match the following related to fuses and their applications.

- (a) Glass cartridge fuse → (i) Emergency protection of electrical circuits
 (b) Kit Kat Fuse → (ii) Power distribution systems
 (c) HRC Fuse → (iii) Electronic equipments
 (d) Fuse wire → (iv) Residential, small scale industrial applications

(1) a – ii, b – iii, c – iv, d – i

(2) a – iii, b – iv, c – ii, d – i

(3) a – iv, b – i, c – ii, d – iii

(4) a – iii, b – ii, c – iv, d – i

✓ ANSWER: (2)

Glass cartridge fuse → (iii) Electronic equipment; Kit Kat Fuse → (iv) Residential/small industrial; HRC Fuse → (ii) Power distribution; Fuse wire → (i) Emergency protection.

If N_1 represents primary windings and N_2 represents secondary windings of transformer, match:

- (a) $N_1 = N_2$ → (i) Auto transformer
 (b) $N_1 < N_2$ → (ii) Isolation transformer
 (c) $N_1 > N_2$ → (iii) Step-up transformer
 (d) Only one winding → (iv) Step-down transformer

(1) a – ii, b – iii, c – iv, d – i

(2) a – i, b – ii, c – iii, d – iv

(3) a – iii, b – ii, c – iv, d – i

(4) a – ii, b – i, c – iii, d – iv

✓ ANSWER: (1)

$N_1 = N_2$ → (ii) Isolation transformer (1:1 ratio); $N_1 < N_2$ → (iii) Step-up (increases voltage); $N_1 > N_2$ → (iv) Step-down (decreases voltage); One winding → (i) Auto transformer.

Identify the properties of ideal transformer.

An ideal transformer is one that has:

- (a) No winding resistance
- (b) No leakage flux
- (c) No iron losses
- (d) High winding resistance

(1) (a), (b) and (c)

(2) (b), (c) and (d)

(3) (c) and (d)

(4) Only (d)

✓ ANSWER: (1)

An ideal transformer has: (a) zero winding resistance, (b) no leakage flux, and (c) no iron losses — meaning 100% efficiency. (d) High winding resistance is NOT a property of an ideal transformer.

A 1000/100V, 20 kVA transformer has 50 turns in the secondary. Calculate the primary turns.

(1) 50

(2) 500

(3) 5

(4) 5000

✓ ANSWER: (2)

$$V_1/V_2 = N_1/N_2$$

$$1000/100 = N_1/50$$

$$N_1 = 50 \times 10 = 500 \text{ turns}$$

The working principle of transformer is based on _____.

- (1) Coulomb's law (2) Faraday's Law
(3) Newton's Law (4) Ohm's Law

✓ ANSWER: (2)

A transformer works on **Faraday's Law of Electromagnetic Induction** — a changing magnetic flux through a coil induces an EMF in the coil.

The capacity of battery is primarily measured in _____

- (1) Voltage (V) (2) Power (W)
(3) Ampere-hours (AH) (4) Current (I)

✓ ANSWER: (3)

Battery capacity is measured in **Ampere-hours (Ah)** — it indicates how much current a battery can deliver over a specific time period.



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